

If you are in the business of being a seer, describing the future to other less-privileged mortals, you are judged on the merits of your predictions.

Helenus, in *The Iliad*, was a different kind of seer. The son of Priam and Hecuba, he was the cleverest man in the Trojan army. It was he who, under torture, told the Achaeans how they would capture Troy (apparently he didn't predict that he himself would be captured). But this is not what distinguished him. Helenus, unlike other seers, was able to predict *the past* with great precision—without having been given any details of it. He predicted backward.

Our problem is not just that we do not know the future, we do not know much of the past either. We badly need someone like Helenus if we are to know history. Let us see how.

The Melting Ice Cube

Consider the following thought experiment borrowed from my friends Aaron Brown and Paul Wilmott:

Operation 1 (the melting ice cube): Imagine an ice cube and consider how it may melt over the next two hours while you play a few rounds of poker with your friends. Try to envision the shape of the resulting puddle.

Operation 2 (where did the water come from?): Consider a puddle of water on the floor. Now try to reconstruct in your mind's eye the shape of the ice cube it may once have been. Note that the puddle may not have necessarily originated from an ice cube.

The second operation is harder. Helenus indeed had to have skills.

The difference between these two processes resides in the following. If you have the right models (and some time on your hands, and nothing better to do) you can predict with great precision how the ice cube will melt—this is a specific engineering problem devoid of complexity, easier than the one involving billiard balls. However, from the pool of water you can build infinite possible ice cubes, if there was in fact an ice cube there at all. The first direction, from the ice cube to the puddle, is called the *forward process*. The second direction, the *backward process*, is much, much more complicated. The forward process is generally used in physics and engineering; the backward process in nonrepeatable, nonexperimental historical approaches.

In a way, the limitations that prevent us from unfrying an egg also prevent us from reverse engineering history.

Now, let me increase the complexity of the forward-backward problem just